

Applied Data Analytics Capstone Report for Analyzing Elite Pitcher Performance with Adjusted Metrics

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for

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1. EXECUTIVE SUMMARY

In this project, I dove more into the careers of five of the greatest pitchers of all time in Major League Baseball history: Roger Clemens, Pedro Martinez, Randy Johnson, Greg Maddux, and Clayton Kershaw. I attempted to gather objective data that could compare their performance over an extended prime window that I adjusted to 162-game seasons, while factoring in and controlling for the pitcher's era and the ballparks that they pitched in throughout their careers. By having my data normalized to the length of each player's best 15 years, I was then able to clean and find the patterns in the data to be directly compared to each other, allowing for the conclusion to be more definitive. I performed an EDA on the potential interactions and correlation of ERA+, Strikeouts (SO), and Wins Above Replacement (WAR), and I found that the results for ERA+ and SO, as well as WAR and SO, from the model of a simple multi-linear regression to be very similar to the results of the EDA. The results showed a strong positive relationship between all three, but it also established which pitchers that succeeded from a velocity-driven style of play and set up the basis of the predictive model done later on.

2. INTRODUCTION

The objective of this study was to analyze the careers of five MLB pitching legends and attempt to determine how these players, who had very different pitching styles, were able to be successful for such long periods of time. The dataset was modified slightly, and the player seasons analyzed were filtered to 15-year stretches in each pitcher's career. The guiding research question for this project was: "How did five elite pitchers with different pitching styles end up being successful in the Major Leagues as seen by their era- and park-adjusted metrics?" The hope was to confirm a hard correlation between the obvious performance statistics, such as ERA+ and SO, and the total value of a player as shown by their WAR.

3. DATA SELECTION APPROACH

The data used in this project was manually scraped from the reputable public baseball statistics website Baseball-Reference.com. I had applied the same information filter to manually scrape the data needed to compare and contrast the careers of the five selected pitchers based on their performance and pitching style. The reason for choosing this data was the need for specific and small sample data that can provide targeted and point-for-point comparison between the elite performers. The information was composed of 75 rows with 16 different columns of player names and pitching statistics, with no missing information and easy to follow preparation procedure. The main variables for this project analysis were Wins Above Replacement (WAR), ERA+, and Strikeouts (SO).

4. METHODS USED

The data was collected and pre-processed into a single dataset. The data was then analyzed using Python programming language with the Pandas library for data analysis. Matplotlib and Seaborn were used for data visualization. The simple multi-linear regression model was used to describe the relationship between the selected independent variables and the dependent variable to provide an initial model of statistical analysis.

5. ANALYSIS AND RESULTS

5.1 Data Analysis Objective(s)

The purpose of the analysis is to predict a pitcher's value (WAR) based on the combination of their pitching effectiveness (ERA+) and their ability to induce strikeouts (SO).

- Dependent Variable: WAR (Wins Above Replacement)
- Independent Variables: ERA+ (Earned Run Average Plus) and SO (Strikeouts)

5.2 Data Cleaning/Editing

The data from the source material was raw and cluttered; it contained more than 30 columns with different statistics. The dataset contained a lot of variables that weren't useful in answering the research question. The data also included career totals and other unnecessary information. The following are the data cleaning/editing processes that were necessary to get to a point where it could be analyzed:

- **Column Selection:** The dataset was filtered only to include the most pertinent variables, dropping all other columns. Only Player, Season, Team, and the pitching WAR, ERA+, SO, FIP, and other key performance indicators were kept for analysis.
- **Data Formatting:** The Player and Season columns were standardized as string and integer data types, respectively. The remaining columns were checked to ensure the data types were appropriate for the analysis, which they were. The dataset did not contain any missing values, which was a significant plus in the cleaning process.
- **Data Standardization:** The most crucial process was standardizing the data by filtering each player's data to only a standard 15-year "prime" window. This standardization ensured the data was fair to compare and more emphasized their best years than to show a large data set for one pitcher whose career was simply longer than the others.

The final cleaned dataset included 75 rows and 16 columns.

5.3 Data Coding

No additional data coding was necessary.

5.4 Types of Variables (Scales of Measurement)

All the variables used in this analysis were quantitative. WAR, ERA+, and SO were all ratio-level variables, as they have a true zero point and allow for meaningful comparison.

5.5 Data Reduction

No rows were removed as there were no invalid or null values in the refined dataset. However, significant data reduction was performed by focusing only on the most informative variables and standardizing the data to the 15-year prime window.

5.6 Data Visualization and Meaning

The EDA showed several useful aspects in histograms, box plots, and heatmaps. Histograms and boxplots indicated that many of the pitching volume statistics (H, ER, HR, etc.) follow a right-skewed distribution, which is typical of counting statistics where a few players will have extremely high values. The boxplots also illustrated the existence of outliers in WAR, SO, and SO/BB, which in most cases represent a season by an individual player (most commonly Randy Johnson or Pedro Martinez).

The most important conclusion from the correlation heatmap is the high positive correlation between WAR and the two key metrics of ERA+ and SO. This high correlation was the primary reason those two variables were selected: they capture that pitchers who allow fewer runs and who strike out batters are more valuable. The negative correlation between ERA+ and the traditional counting statistics is unsurprisingly high, as ERA+ is a measure of performance with higher values being better.

5.7 Sample Size (Training and Test Data)

After the above filtering criteria were applied, I was left with 75 total observations across the five pitchers. In each pitcher's 15 data points, I have, essentially, a standardized 15-year prime period against which to compare.

Before attempting to build any sort of model on this dataset, I created a training set and a test set. I split the data 70/30 using a random selection of one pitcher's data; there were 52

observations in my training set, and 23 in my test set. My reasoning in doing this was to allow the model enough observations to draw strong conclusions about the relationships between different variables and the outcome. But by holding back enough observations (30% in this case), I could be reasonably sure that when I calculated the accuracy measures (Mean Absolute Error, R-squared, etc.) of my model, I was getting a fair assessment of how it would perform on out-of-sample data.

5.8 Analysis Bias

The most important source of potential bias in this analysis is selection bias. This sample was not random, because only the top five pitchers were hand-selected, and so the results do not generalize to the population of MLB pitchers.

6. CONCLUSION

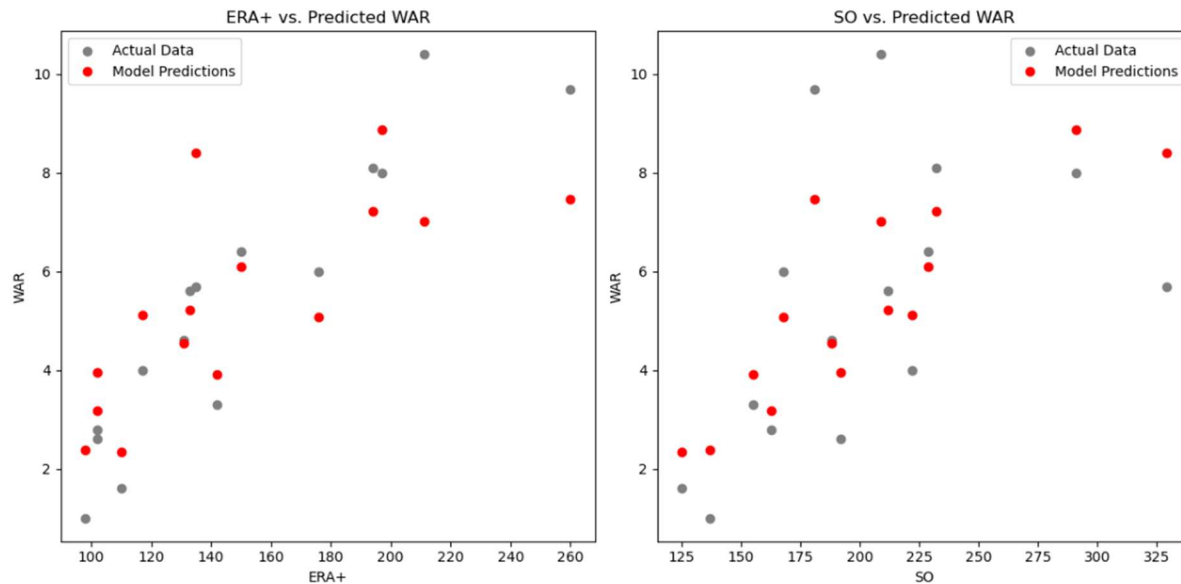
The analysis confirms a strong relationship between ERA+ and Strikeouts and a pitcher's overall value as measured by WAR. It confirmed that even among a group of elite athletes, the ability to prevent runs and generate strikeouts remains a central driver of success. The findings from the exploratory data analysis and the simple multi-linear regression model proved that these metrics are central to a pitcher's success in a way that transcends era, team, or specific pitching style.

6.1 Model Summary

A simple multi-linear regression model was used to predict WAR based on ERA+ and Strikeouts. The model confirmed that these variables were significant predictors of a pitcher's value. The model suggested that as ERA+ and Strikeouts increase, so does a pitcher's WAR.

6.1.1 Actual vs. Predicted Values

The pair plots below show the model-predicted WAR values in red plotted against the dataset's WAR values with ERA+ and Strikeouts (SO) as independent variables, respectively. The gray dots are the datapoints from the original dataset. The red dots show the predicted values the model made for each of those data points.



ERA+ vs. Predicted WAR: From the plot, we can see that the model has fit the general increase of predicted values(red) as ERA+ increases. The model is accurate for lower ERA+ values that are near the 100 mark because they have predicted values at the lower end, like the rest of the data. As ERA+ increases, the red dots increase in value and remain somewhat linear, but for a few points with around 5 WAR, there are both gray and red dots, showing that the actual data and the predicted data are different. This plot is a good example of the model's overfit as it matches the data trends but still has many points that have room for improvement.

SO vs. Predicted WAR: The predicted WAR (red) is in line with the general upward trend of the rest of the data (gray). Like the other plot, as SO increases, so does the predicted WAR of a pitcher. The predicted values increase as the amount of strikeouts increases, and then

there is some decline in the middle, but this plot does not have as many WAR with high strikeouts, as some of the largest SO in the data also have lower predicted WAR than the rest of the data.

6.2 Coefficient Interpretation

The coefficients for the linear regression model can be interpreted as the effect of each independent variable on the pitcher's WAR. The intercept for the model was -3.64. This is the expected value of the pitcher's WAR when ERA+ and Strikeouts are equal to 0. The two coefficients were as follows:

- ERA+: 0.024
- SO: 0.026

The coefficient for ERA+ was 0.024. This meant that a one-point increase in a pitcher's ERA+ was associated with an increase in WAR of 0.024, all other variables being equal. Since a higher ERA+ indicated a better performance, it makes sense that a higher ERA+ would be associated with a higher WAR. The coefficient for Strikeouts was also positive at 0.0267. This meant that a one unit increase in Strikeouts was associated with an increase in WAR of 0.0267, all other variables being equal. I could then conclude that run prevention and strikeout ability are both separate, positive contributors to pitcher value.

6.3 Evaluation Metrics

Evaluation metrics are used to quantify the performance of a machine learning model.

- **Mean Absolute Error (MAE):** The MAE would represent the average difference between the model's predicted WAR and the actual WAR.
- **Mean Square Error (MSE):** The MSE would penalize larger prediction errors more heavily.

- **Root Mean Square Error (RMSE):** The RMSE would be on the same scale as the WAR variable, making it easier to interpret.
- **R-Squared Score (R^2):** The R-squared score would show the proportion of the variance in WAR that is explained by the independent variables. A high R^2 value would indicate that the model is a strong fit for the data.

6.3.1 Mean Absolute Error Result

The Mean Absolute Error (MAE) is 1.15. This value states that the model's predicted and actual WAR values differ by, on average, 1.15 wins. This result is good, as the margin of error, on average, was quite low compared to the dependent variable, WAR.

6.3.2 Root Mean Square Error Result

The Root Mean Square Error (RMSE) is 1.47. This value tells us our errors from our true values, in the same units as the dependent variable (WAR), which is preferred. The value also states that, on average, our predicted values and true values differed by 1.47 wins. This value is slightly larger than our MAE value because there were a few big individual errors, which were squared and thus, made bigger in this metric.

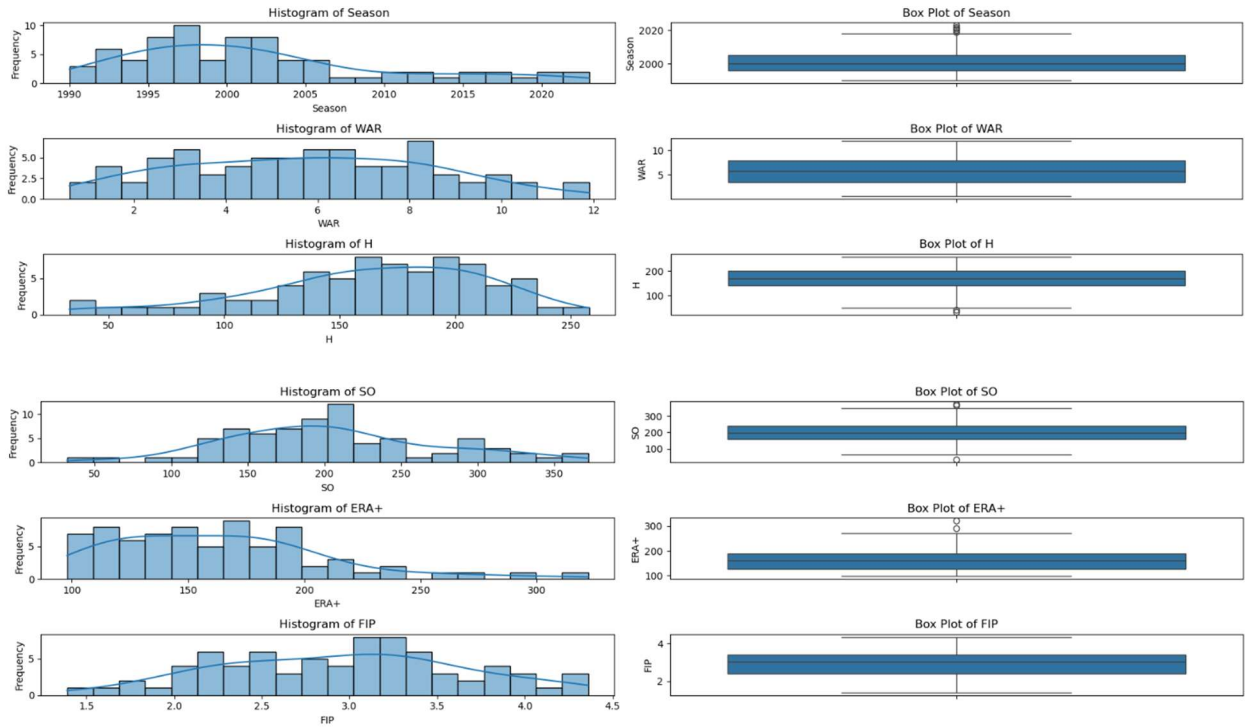
6.3.2 R-Square Score Result

The R-Squared Score is 0.717. This value indicates how much of the independent variable explains the changes in the dependent variable, which is approximately 71.7%. This is an extremely good result, and ERA+ and Strikeouts are great at explaining a pitcher's WAR

APPENDICES

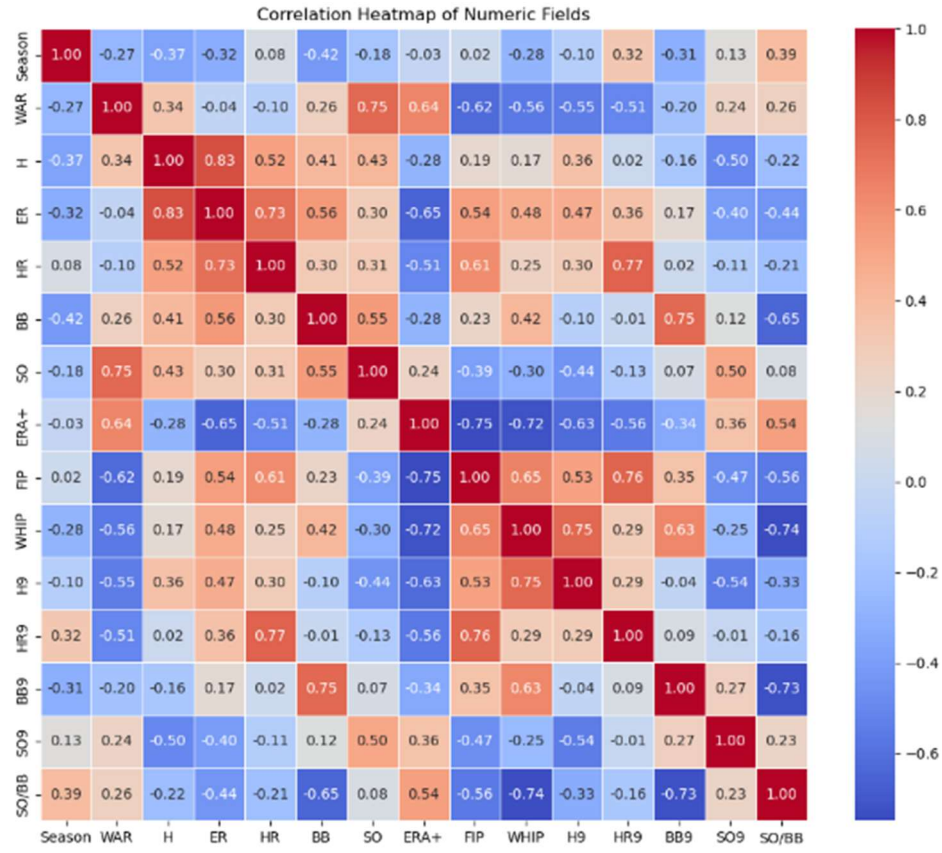
Appendix A

Independent and Dependent Variable Distribution Graph



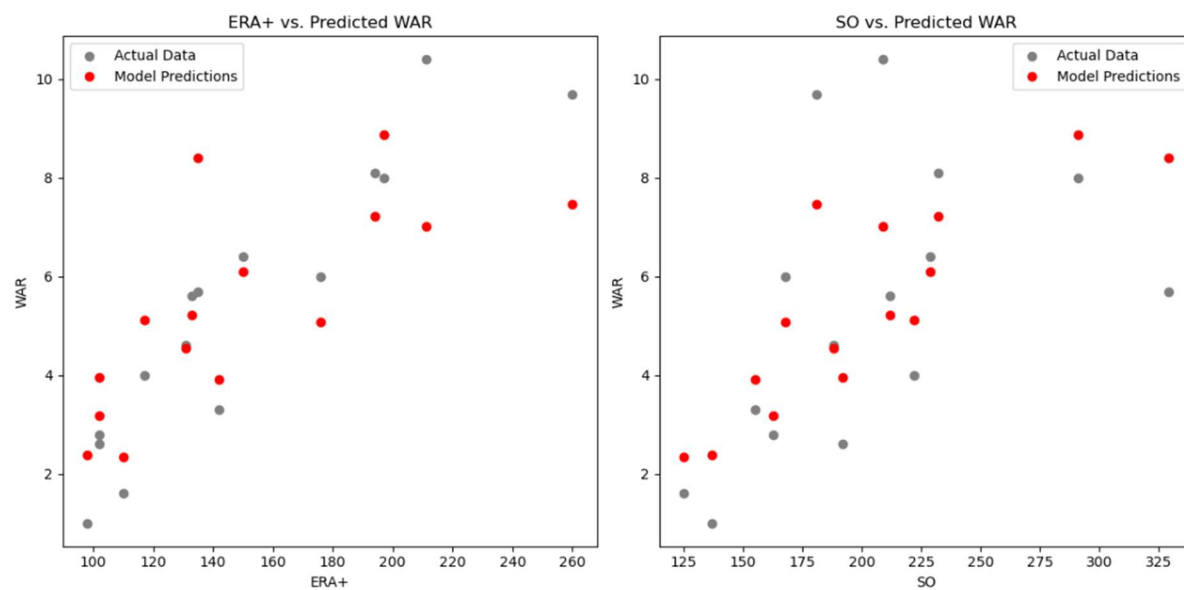
Appendix B

HeatMap Plot Variable Relationships



Appendix C

Pair Plot Variable Relationships



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